5.5

Analysis of Accounts

Cambridge IGCSE Business

Profitability	Liquidity
GPM	Profit Margin
ROCE	Current Ratio
Acid Test	Users & Limits

Analysis of Accounts

Ratio analysis uses figures from the profit and loss account and balance sheet to measure financial performance. It is one of the most calculation-heavy topics in IGCSE Business — and one of the most rewarding to master.

- Calculate gross profit margin, profit margin and ROCE
- Calculate and interpret the current ratio and acid test
- Explain how different users make decisions from ratios
- Analyse trends and identify limitations of ratio analysis

Profitability & Liquidity

Two different measures of financial health

Profitability and Liquidity – Two Different Concepts

Ratio analysis compares figures from the financial statements to give a meaningful picture of performance. Profitability and liquidity measure completely different things.

Profitability vs Liquidity — Two Different Measures

PROFITABILITY

Definition:	How well a business generates profit relative to its size or investment
Question:	"Is this business making enough profit for its size?"
Ratios:	Gross Profit Margin · Profit Margin · ROCE
Used by:	Investors, managers, shareholders

High profitability = efficient, generating good returns on investment

LIQUIDITY

Definition:	How easily a business can pay its short-term debts
Question:	"Can this business pay its bills right now?"
Ratios:	Current Ratio · Acid Test Ratio
Used by:	Banks, suppliers, creditors

High liquidity = can meet obligations without selling assets

Critical distinction: a business can be PROFITABLE but ILLIQUID — and fail. It can be LIQUID but UNPROFITABLE — and survive short-term but not long-term. Always treat them separately.

The Five Ratios — Formula Reference Card

Gross Profit Margin	Profit Margin	ROCE	Current Ratio	Acid Test Ratio
Formula: Gross Profit ÷ Revenue × 100	Formula: Net Profit ÷ Revenue × 100	Formula: Net Profit ÷ Capital Employed × 100	Formula: Current Assets ÷ Current Liabilities	Formula: (Current Assets − Inventory) ÷ Current Liabilities
Café example: e.g. £57,000 ÷ £95,000 × 100 = 60%	Café example: e.g. £17,400 ÷ £95,000 × 100 = 18.3%	Café example: e.g. £17,400 ÷ £67,500 × 100 = 25.8%	Café example: e.g. £14,500 ÷ £7,000 = 2.07:1	Café example: e.g. (£14,500−£4,500) ÷ £7,000 = 1.43:1
Benchmark: Higher is better · com pare year-on-year	Benchmark: Higher is better · ind ustry benchmark varies	Benchmark: Higher is better · com pare to interest rates	Benchmark: Ideal: 1.5:1 to 2:1	Benchmark: Ideal: 1:1 or above
For every £1 of revenue,	For every £1 of revenue,	Every £100 invested gene	For every £1 owed, the b	Excludes inventory — str

Exam tip: You MUST know all five ratio formulas. The five are: Gross Profit Margin, Profit Margin, ROCE, Current Ratio, AcidTest. Learn the formula, the worked example and the ideal benchmark for each.

Profitability Ratios

Gross Profit Margin · Profit Margin · ROCE

Gross Profit Margin (GPM) – Formula, Calculation and Interpretation

The gross profit margin shows what percentage of each pound of revenue is left after paying the direct costs of producing or buying the goods sold.

Gross Profit Margin

$$\frac{\text{Gross Profit}}{\text{Revenue}} \times 100$$

Riverside Café: £57,000 ÷ £95,000 × 100
= 60%

For every £1 of revenue, 60p is gross profit

Higher = better · rising GPM = cost of sales falling or prices rising

What does the GPM tell us?

- A GPM of 60% means 60p of every £1 of revenue is gross profit
- A rising GPM is a positive sign — the business is becoming more efficient at managing its direct costs
- A falling GPM is a warning — cost of sales rising faster than revenue, or prices cut too deeply
- Compare to the previous year AND to industry competitors to judge whether 60% is strong or weak

How to improve the Gross Profit Margin:

Raise selling price	If demand is inelastic, higher prices increase revenue without losing many customers
Reduce cost of sales	Negotiate better deals with suppliers or switch to cheaper raw materials
Change product mix	Focus on higher-margin products — drop or reduce lower-margin lines
Improve efficiency	Reduce waste in production to lower the cost of each unit produced

Exam tip: GPM only includes COST OF SALES (direct costs). If GPM is high but profit margin is low, the business has high operating expenses (rent, wages, admin). This is a key exam distinction.

Profit Margin (Net Profit Margin) – Formula, Calculation and Interpretation

The profit margin shows what percentage of each pound of revenue is left after ALL costs are deducted — direct and indirect. It is the truest measure of overall profitability.

Profit Margin (Net Profit Margin)

$$\frac{\text{Net Profit}}{\text{Revenue}} \times 100$$

Riverside Café: £17,400 ÷ £95,000 × 100
= 18.3%
For every £1 of revenue, 18.3p is net profit

Higher = better · falling margin = expenses growing faster than revenue

What does the Profit Margin tell us?

- 18.3% means the café keeps 18.3p for every £1 of revenue earned
- Includes ALL costs — rent, wages, marketing, depreciation
- A falling profit margin means expenses are rising faster than revenue
- If GPM is high but profit margin is low — overhead costs are the problem to tackle

Linking GPM and Profit Margin — what the gap tells you:

GPM = 60%

Net Margin = 18.3%

Gap = 41.7% — this is the EXPENSE RATIO (cost of overheads as % of revenue)

Exam tip: Profit Margin includes ALL costs — both direct and indirect. GPM only includes direct costs. The bigger the gap between GPM and profit margin, the higher the overhead burden.

Return on Capital Employed (ROCE) – Formula, Calculation and Interpretation

ROCE measures how much net profit a business generates for every £100 of capital invested in it. It tells investors and managers whether the business is using its funds efficiently.

Return on Capital Employed (ROCE)

$$\frac{\text{Net Profit}}{\text{Capital Employed}} \times 100$$

Riverside Café: £17,400 ÷ £67,500 × 100
= 25.8%
Every £100 invested generates £25.80 profit

Higher = better · compare to bank interest rate as opportunity cost

What does ROCE tell us?

- 25.8% means every £100 invested generates £25.80 of net profit
- Compare to the bank interest rate — if ROCE > interest rate, the investment is worthwhile
- Falling ROCE = capital growing faster than profit — assets may not be earning their keep
- Always compare to competitors — a 10% ROCE is great in retail, poor in tech

How to improve ROCE:

Increase the Numerator (Net Profit)

- Raise prices (if demand is inelastic)
- Cut costs — reduce overheads, negotiate with suppliers
- Grow revenue — attract more customers, expand range

Reduce the Denominator (Capital Employed)

- Repay long-term loans to reduce capital employed
- Sell unused non-current assets that are not earning returns
- Avoid unnecessary capital investment in low-return assets

Exam tip: ROCE is the most comprehensive profitability ratio. It links the income statement AND the balance sheet. If ROCE <bank interest rate, shareholders would be better off putting money in a savings account.

Liquidity Ratios

Current Ratio · Acid Test Ratio · Benchmarks

Current Ratio – Formula, Calculation and Interpretation

The current ratio measures whether a business has enough current assets to pay its current liabilities. It is the primary test of short-term liquidity.

Interpreting 2.07:1 for Riverside Café:

- ✅ 2.07:1 is within the ideal range of 1.5:1 to 2:1 — the business is liquid
- ✅ For every £1 of short-term debt, the café holds £2.07 of short-term assets
- ❌ A ratio below 1:1 means the business CANNOT pay its immediate debts - crisis
- ❌ Above 3:1 suggests too much cash tied up unproductively in current assets
- ★ The ideal range balances safety (enough liquidity) with efficiency (no idle cash)

What is included in each side of the ratio:

Current Assets	Examples	Current Liabilities	Examples
Inventory (stock)	Raw materials, finished goods ready to sell	**Trade payables**	Money owed to suppliers for goods already received
Trade receivables	Money owed TO the business by customers	**Bank overdraft**	Short-term borrowing from the bank
Cash and bank balance	Physical cash + money in bank accounts	**Tax payable**	Corporation tax owed to government this year

Exam tip: The ideal current ratio is 1.5:1 to 2:1. Always state the benchmark in your answer. Never just say '2.07:1 is good' — say '2.07:1 is within the ideal range of 1.5:1 to 2:1'.

Acid Test Ratio – Formula, Calculation and Interpretation

The acid test ratio is a stricter test of liquidity that removes inventory from current assets. It tests whether the business can pay its debts WITHOUT needing to sell its stock first.

Acid Test Ratio

$$\frac{\text{Current Assets} - \text{Inventory}}{\text{Current Liabilities}}$$

Riverside Café: (£14,500 – £4,500) ÷ £7,000
= 1.43:1 Excluding inventory, £1.43 available per £1 owed

Why stricter than the current ratio:
Inventory may be hard to sell quickly — the acid test removes it
Ideal benchmark: 1:1 or above · Below 0.5:1 = serious concern

Why remove inventory?

- Inventory may take a long time to sell — it is the least liquid current asset
- In a crisis, stock may have to be sold at a loss to raise cash quickly
- The acid test gives a more realistic picture of IMMEDIATE liquidity
- A ratio of 1.43:1 means: even without selling stock, the café can pay its debts
- Ideal benchmark: 1:1 or above. Below 0.5:1 = serious short-term risk

Comparing Current Ratio and Acid Test — what the gap tells you:

Current Ratio	**2.07:1**	Includes inventory (£4,500) in current assets
Acid Test Ratio	**1.43:1**	Excludes inventory — only cash and debtors counted

The gap (2.07 – 1.43 = 0.64) represents the impact of inventory. A large gap = business is heavily reliant on its stock — a risk if stock cannot be sold quickly.

Exam tip: Current ratio and acid test should both be calculated. If the current ratio looks fine but the acid test is below 1:1, the business has a hidden liquidity risk — its stock may be difficult to sell.

Trends, Users & Limitations

Comparing over time · Who uses ratios · What they can't tell you

Trend Analysis – Comparing Ratios Over Time

A single ratio in isolation tells you very little. The real value of ratio analysis comes from comparing results over multiple years, against competitors or against industry benchmarks.

Trend Analysis — Riverside Café (3-Year Comparison)

Ratio	2023	2024	2025	Trend	Interpretation
GPM	55%	58%	60%	▲	Rising — cost of sales falling relative to revenue · supplier prices negotiated down
Profit Margin	14%	16%	18.3%	▲	Rising — expenses controlled well · efficiency improving year on year
ROCE	19%	22%	25.8%	▲	Improving — generating more profit per £1 of capital invested
Current Ratio	2.4:1	2.2:1	2.07:1	▼	Falling but still in ideal range (1.5–2:1). Worth monitoring.
Acid Test	1.8:1	1.6:1	1.43:1	▼	Falling — still healthy above 1:1 but inventory growing · check stock levels

Trend analysis is more useful than a single year — it shows the direction of performance over time

Three ways to use ratio analysis effectively:

Time-series comparison	Compare the SAME ratio over 3–5 years for the SAME business. Shows whether performance is improving or declining. Most common in the exam.
Inter-firm comparison	Compare ratios against a COMPETITOR or INDUSTRY AVERAGE. Shows whether the business is outperforming or underperforming its sector peers.
Against benchmarks	Compare against TARGET ratios. Current ratio: ideal 1.5:1 to 2:1. Acid test: 1:1 or above. ROCE should beat the bank interest rate.

Exam tip: Always refer to a benchmark when interpreting ratios. Never just state the number. Say 'the current ratio is 2.07:1 which is within the ideal range of 1.5:1 to 2:1 suggesting the business is liquid'.

Users of Accounts – Who Uses Financial Information and Why

Different stakeholders use the accounts and ratios to make very different decisions. The same set of financial statements can tell many different stories depending on who is reading them.

Who Uses Accounts and What Do They Look For?

Shareholders / Owners

Key question:
"Is the business profitable enough to justify my inve"

They look at:
- Profitability ratios — GPM, prof it margin, ROCE
- Dividend payments and retained p rofit
- Year-on-year trend in net profit

Managers

Key question:
"How efficiently is the bus iness performing?"

They look at:
- All ratios — to identify areas n eeding improvement
- Compare against competitors and budget targets
- Use to make pricing and cost dec isions

Banks / Lenders

Key question:
"Can this business repay a loan?"

They look at:
- Liquidity ratios — current ratio , acid test
- Profitability — can they service debt from profits?
- Trend over time — improving or d eteriorating?

Suppliers

Key question:
"Will this business pay my invoices on time?"

They look at:
- Current ratio and acid test rati o
- How quickly does it pay existing creditors?
- Is it financially stable enough to keep ordering?

How each user makes a decision from the accounts:

User	Key ratio	Decision they make
Shareholder	ROCE vs bank rate	Invest more / hold shares / sell — is return beating alternatives?
Bank manager	Current ratio + acid test	Approve / decline loan — can business repay its debts?
Supplier	Current ratio	Extend trade credit / demand upfront payment
Manager	All ratios + trend	Set targets, cut costs, raise prices, invest in assets
Employee	Profit margin trend	Negotiate pay rise / assess job security
Government	Profit — revenue	Calculate tax liability and check compliance

Exam tip: When asked 'how might a bank use ratio analysis?' — name the specific ratios (current ratio, acid test), explain what they show AND state the decision the bank would make.

Limitations of Ratio Analysis – What Ratios Cannot Tell You

Ratio analysis is a powerful tool but it has significant limitations. A good exam answer will use ratios to support conclusions but also acknowledge what they cannot reveal.

Limitations of Ratio Analysis

- Based on HISTORICAL data — ratios reflect the past, not the future
- Different accounting methods make comparisons between firms unreliable
- Ratios do not capture qualitative factors — staff morale, brand reputation, customer satisfaction
- Seasonal businesses have very different ratios at different points in the year
- Ratios alone cannot explain WHY performance has changed — only that it has
- Small changes in figures can make ratios look very different — ratios can be manipulated
- No single 'ideal' benchmark works across all industries — context always matters

Use ratios alongside:

Qualitative information	Customer feedback, employee satisfaction surveys, brand perception research
Competitor data	Industry benchmarks and inter-firm comparisons for meaningful context
Management commentary	The story behind the numbers — why did costs rise? Why did revenue fall?
Cash flow forecasts	Ratios are backward-looking — cash flow forecasts look forward
Non-financial data	Market share, number of new customers, staff turnover, product quality scores

"Ratios are a starting point, not a conclusion. They raise questions — it is the job of the manager to answer them."

Exam tip: When evaluating, always state ONE limitation of ratio analysis. For example: 'However, these ratios are based on historical data and may not reflect the current trading position of the business'.

Key Revision Checklist – 5.5 Analysis of Accounts

GPM	(Gross Profit ÷ Revenue) × 100. Riverside Café: 60%. Higher = better. Only includes direct costs.
Profit Margin	(Net Profit ÷ Revenue) × 100. Café: 18.3%. Includes ALL costs. If GPM high but PM low → high overheads.
ROCE	(Net Profit ÷ Capital Employed) × 100. Café: 25.8%. Compare to bank rate. Higher = better use of funds.
Current Ratio	Current Assets ÷ Current Liabilities. Café: 2.07:1. Ideal: 1.5:1 to 2:1. Below 1:1 = danger.
Acid Test	(Current Assets – Inventory) ÷ Current Liabilities. Café: 1.43:1. Ideal: above 1:1. Stricter than CR.
Trend analysis	Compare ratios over 3+ years. Falling ratios are a warning sign. Rising ratios signal improving performance.
Shareholders	Use ROCE, profit margin, dividend data to decide whether to invest more or sell their shares
Banks	Use current ratio and acid test to decide whether to lend money or approve an overdraft facility
Limitations	Historical data only · accounting methods differ · no qualitative info · seasonality · can be manipulated